

Day 9: STTP on Ethical Hacking and Cyber Forensics @IIITK

Multimedia Forensics

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1 Introduction to Multimedia Forensics

Definition: Scientific analysis, integrity authentication, and investigation of digital multimedia evidence (images, videos, audio, documents).

2 The Challenge: Fake and Tampered Media

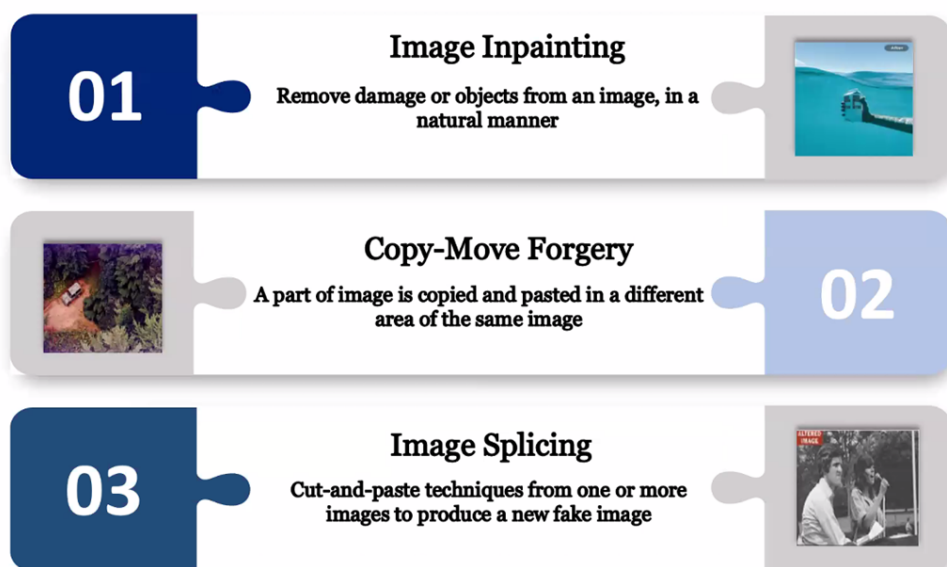
Digital tools enable easy manipulation, leading to:

- Deepfakes (AI-generated realistic fake media).
- Fake News (manipulated images/videos for misinformation).
- Tampered E-documents (altered digital records).
- Image Tampering/Forgery.



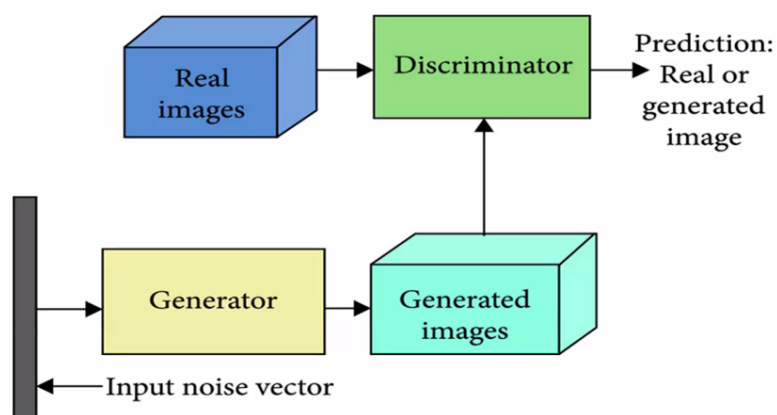
3 Types of Image Forgeries

1. **Image Inpainting:** Removing objects/damage seamlessly.
2. **Copy-Move Forgery:** Copying and pasting within the same image.
3. **Image Splicing:** Combining parts from different images.



4 AI-Generated Fake Images & GANs

AI, particularly Generative Adversarial Networks (GANs), can create hyper-realistic fake images (e.g., thispersondoesnotexist.com).



GANs (Simplified ML): Two neural networks, a **Generator** (creates fakes) and a **Discriminator** (spots fakes), compete and improve, resulting in highly realistic synthetic media.

5 Key Forensic Questions

- **Source identification:** Where did the data originate?
- **Integrity verification/tampering detection:** Has the data been altered?

6 Digital Fingerprints

Unique traces left by devices or processes:

- **In-camera:** From lens, sensor, camera software.
- **Out-camera:** From editing, compression after capture.
- **Scene (geometric):** From lighting, reflections in the scene.

7 Source Identification Levels

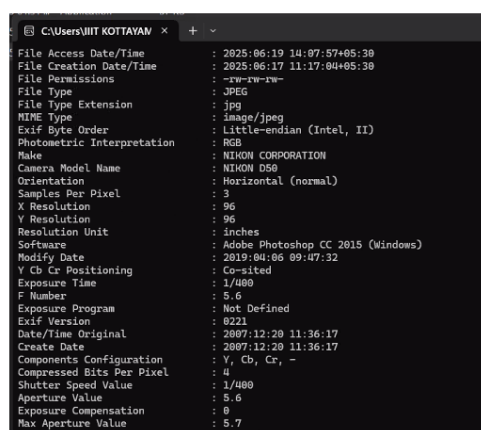
1. **Class Identification:** Device type (Scanner, Smartphone, etc.).
2. **Model/Brand Identification:** Specific model (Nikon D50). Uses lens distortion, CFA patterns.
3. **Device-Level Identification:** The exact device.

Photo Response Non-Uniformity (PRNU): Sensor imperfections create a unique noise pattern (fingerprint). **ML for PRNU:** ML models learn these unique PRNU "fingerprints" from known cameras to match a questioned image to a specific device.

8 Forensic Analysis Techniques

8.1 Metadata Analysis

Examining embedded data (EXIF: date, time, camera, software). Tools like **ExifTool**. **ML for Metadata:** Detects anomalies in metadata patterns that suggest tampering.



```

C:\Users\JIT KOTTAYAM x + -
File Access Date/Time       : 2025:06:19 14:07:57+05:30
File Creation Date/Time    : 2025:06:17 11:17:04+05:30
File Permissions           : -rw-rw-rw-
File Type                  : JPEG
File Type Extension        : jpg
MIME Type                  : image/jpeg
Exif Byte Order             : Little-endian (Intel, II)
Photometric Interpretation  : RGB
Make                       : NIKON CORPORATION
Camera Model Name          : NIKON D50
Orientation                 : Horizontal (normal)
Samples Per Pixel           : 3
X Resolution                : 96
Y Resolution                : 96
Resolution Unit             : inches
Software                   : Adobe Photoshop CC 2015 (Windows)
Modify Date                 : 2019:04:06 09:47:32
Y Cb Cr Positioning        : Co-sited
Exposure Time               : 1/400
F Number                   : 5.6
Exposure Program            : Not Defined
Exif Version                : 0221
Date/Time Original         : 2007:12:20 11:36:17
Create Date                 : 2007:12:20 11:36:17
Components Configuration   : Y, Cb, Cr, -
Compressed Bits Per Pixel   : 4
Shutter Speed Value         : 1/400
Aperture Value              : 5.6
Exposure Compensation      : 0
Max Aperture Value         : 5.7
  
```

8.2 Compression Analysis

Compression artifacts (e.g., from JPEG) can reveal manipulation or multiple saves. **ML for Compression:** CNNs identify specific compression artifacts, history, and inconsistencies indicating splicing.

8.3 Image Analysis Techniques

- **Clone Detection:** Identifying duplicated regions (similar textures, noise).
- **Splicing Detection:** Analyzing inconsistencies (lighting, noise, edges) when images are combined.

ML (CNNs) for Tampering: Convolutional Neural Networks (CNNs) act like "smart filters," trained to automatically spot subtle visual cues of tampering (e.g., inconsistent lighting, noise, edges) that humans might miss.

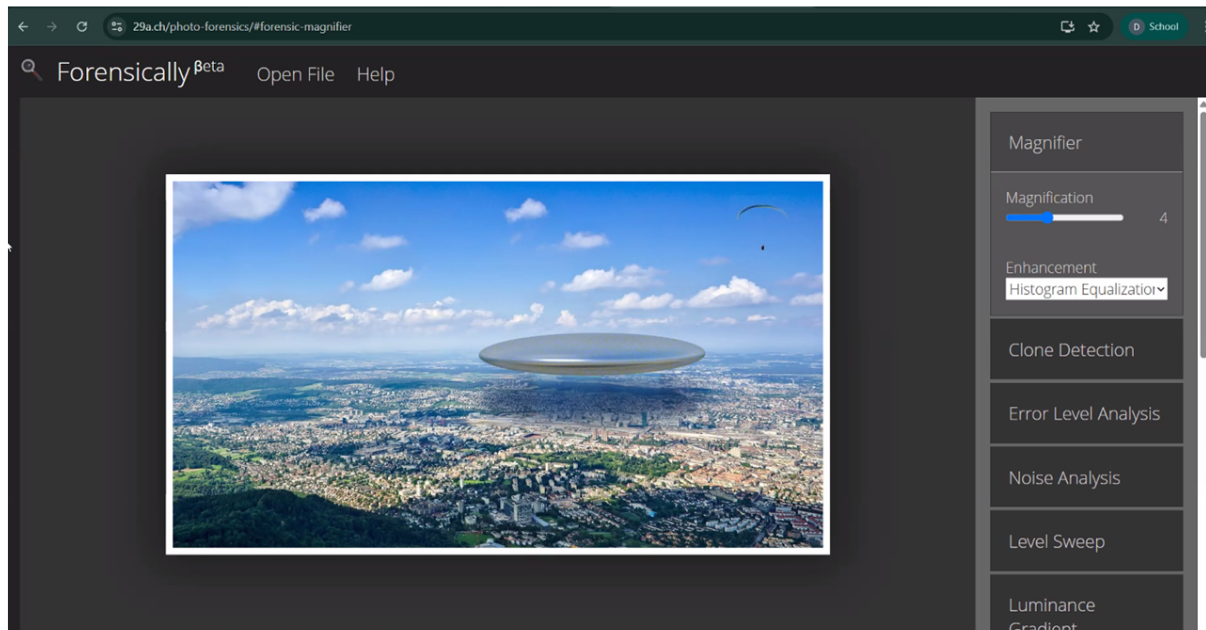


8.4 Other Techniques

- **Motion Deblurring:** Sharpening motion-blurred images.
- **Digital Signatures/Watermarks:** Verifying authenticity through embedded signatures.

9 Tools for Multimedia Forensics

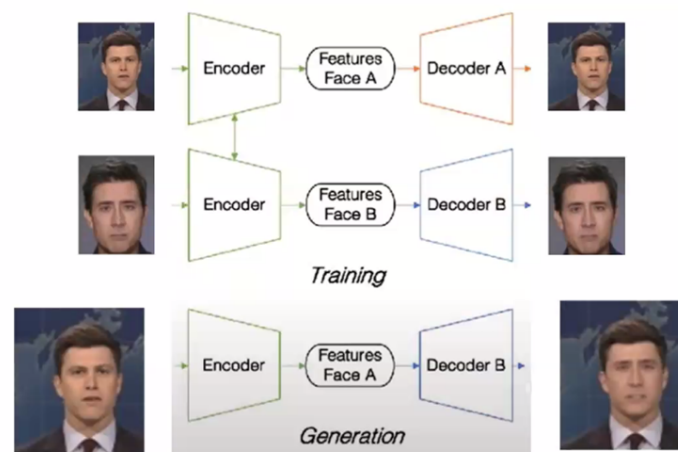
- **ExifTool:** For metadata extraction.
- **FotoForensics:** Known for Error Level Analysis (ELA) to see differing compression levels.
- **Forensically (29a.ch):** Online suite (ELA, clone detection, noise analysis).



10 Deepfake Forensics

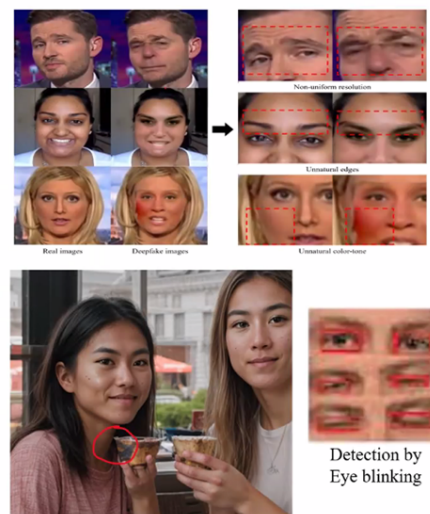
10.1 Deepfake Creation

Often uses autoencoders (ML): an **Encoder** captures facial features, and a different person's **Decoder** reconstructs the face with those features.



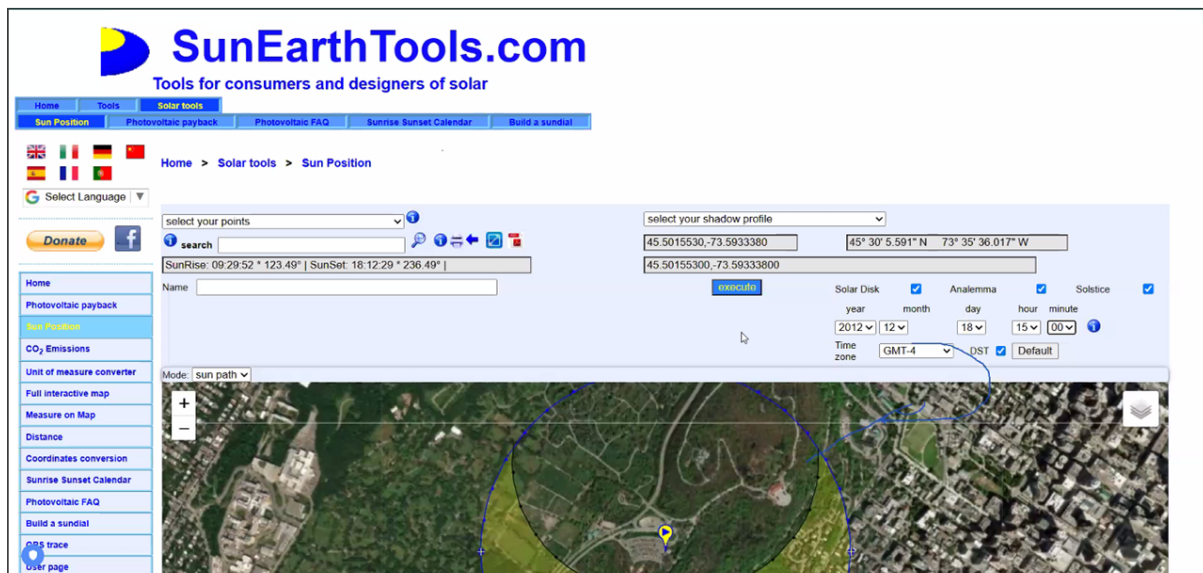
10.2 Deepfake Detection

Challenging. Methods: facial/behavioral analysis, audio checks, context. **ML for Detection:** CNNs analyze frames for visual artifacts (unnatural skin, lighting). RNNs analyze video sequences for inconsistent motion or expressions.



11 Shadow Analysis

Verifying image authenticity by checking shadow consistency with purported time/location using tools like SunEarthTools.com. Inconsistent shadows suggest tampering.



12 Conclusion

Multimedia forensics is vital in combating digital deception. As forgery techniques (especially AI-driven) advance, forensic methods, including robust ML approaches, must evolve to ensure the integrity of digital evidence.